

STUDENT RECORDS MANAGEMENT SYSTEM REPORT

STD NAME: BABIRYE SANDRA

REG NO: 23/U/07270/EVE

1. Introduction

The Student Record Management System is a menu-driven Python application developed to manage student information efficiently. The system allows users to add, view, search, update, and delete student records. It uses a CSV file for storing basic student information and a JSON file for storing additional details such as address, contact number, and program.

The system was developed using Python concepts including file handling, functions, exception handling, data structures, JSON and CSV processing, and logging.

2. Program Design

The application follows a modular design approach where different operations are separated into individual functions. The main menu provides users with options to perform different student management tasks.

The system uses two storage files:

students.csv stores Registration number, Student name, Age and Gender.

students.json stores Address, Contact number and Program.

The main components include:

File initialization.

Student management functions.

Logging system.

3. Key Functions

initialize_files(): Creates the required CSV and JSON files.

add_student(): Validates and saves student records to CSV and JSON.

view_students(): Displays student information from both files.

search_student(): Searches students using the registration number.

update_student(): Updates student information in both files.

delete_student(): Deletes student information from both files.

4. Exception Handling Strategy

The program uses try, except and finally blocks to handle invalid user input, missing files, corrupted JSON files and unexpected runtime errors.

Two custom exceptions were implemented:

- StudentExistsError – raised when a duplicate registration number is entered.
- StudentNotFoundError – raised when a student record cannot be found.

The logging module records successful operations, user activities and system errors in student_system.log.

5. Testing Results

The following tests were carried out successfully:

- Add new student.
- Add duplicate registration number.
- View all students.
- Search existing student.
- Search non-existing student.
- Update student details.
- Delete student record.
- Invalid registration number validation.
- Corrupted JSON file handling without crashing the application.

6. Conclusion

The Student Record Management System successfully demonstrates Python file handling, functions, exception handling, custom exceptions, input validation and logging. The application provides an efficient and reliable way of managing student records while maintaining data integrity.